

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Artificial Intelligence

Semester: Spring

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What do you mean by AI? Why it is important? Explain the objectives of AI with examples. 7
- b) Define rational agent? Explain the Learning agent with block diagram. 8
2. a) Define the problem as state space search. Write the various steps involved in solving water-jug problem and give the solution for it. 8
- b) Write down the searching steps. Differentiate between uninformed search and informed search. 7
3. a) Explain Minmax algorithm with an example. 7
- b) Michele is a superstar. All superstars are rich. Rich mans have fast cars. Fast cars consume a lot of petrol. Prove that Michele' car consumes a lot of petrol using resolution 8
4. a) Discuss the importance of knowledge based systems and explain the representation of knowledge. 8
- b) Explain the syntax and semantics for first order predicate logic. 7
5. a) What is Machine learning? Explain the learning by analogy and explation based learning. 8
- b) Define fuzzy learning. Explain reinforcement learning. 7
6. a) Explain about Back Propagation Algorithm in Neural network. 8

OR

Define Perceptron. Illustrate one of the neural network training example.

- b) Why Natural language understanding is difficult? And also explain its application. 7
- 7 Write short notes on: (Any two) 2×5
 - a) Adaline Network
 - b) Boltzmann Machine
 - c) Reinforcement Learning.